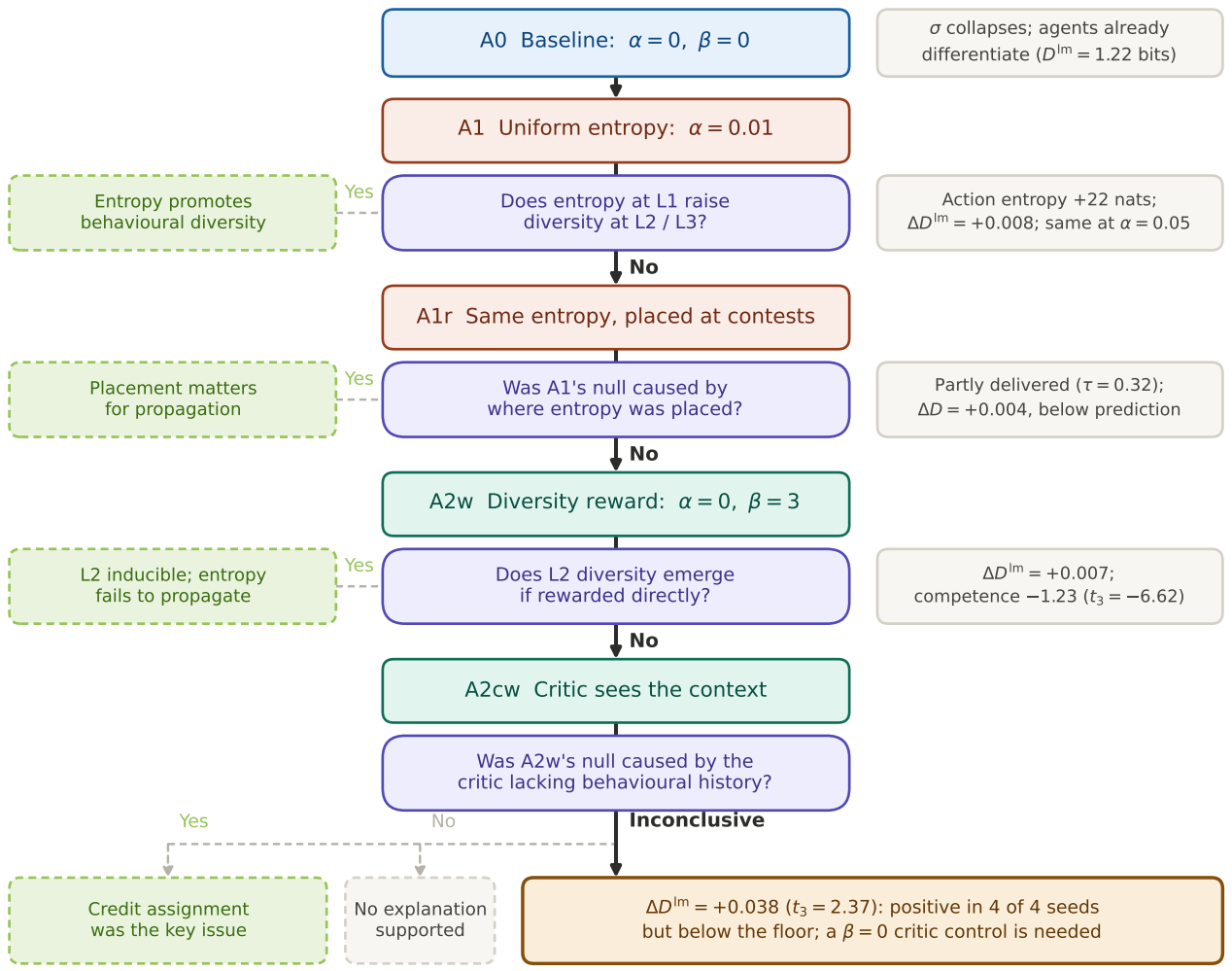
A0 Baseline: $\alpha = 0,\ \beta = 0$
σ collapses; agents already differentiate ($D^{\text{lm}} = 1.22$ bits)
A1 Uniform entropy: $\alpha = 0.01$
Entropy promotes behavioural diversity
Yes
Does entropy at L1 raise diversity at L2 / L3?
Action entropy +22 nats; $\Delta D^{\text{lm}} = +0.008$; same at $\alpha = 0.05$
No
A1r Same entropy, placed at contests
Placement matters for propagation
Yes
Was A1's null caused by where entropy was placed?
Partly delivered ($\tau = 0.32$); $\Delta D = +0.004$, below prediction
No
A2w Diversity reward: $\alpha = 0,\ \beta = 3$
L2 inducible; entropy fails to propagate
Yes
Does L2 diversity emerge if rewarded directly?
$\Delta D^{\text{lm}} = +0.007$; competence -1.23 ($t_3 = -6.62$)
No
A2cw Critic sees the context
Was A2w's null caused by the critic lacking behavioural history?
Yes
No
Inconclusive
Credit assignment was the key issue
No explanation supported
$\Delta D^{\text{lm}} = +0.038$ ($t_3 = 2.37$): positive in 4 of 4 seeds but below the floor; a $\beta = 0$ critic control is needed